Adaptive Machine Learning Models for Predicting Customer Behaviour in Dynamic Market Environments to Support Personalised Business Strategy Development

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Table of Contents

1. Introduction....3

2. Background and Rationale....3

3. Research Aim....4

4. Research Objectives....4

5. Research Questions....5

6. Literature Review....5

6.1 Evolution of Customer Behaviour Analytics....5

6.2 Adaptive Machine Learning Techniques....5

6.3 Explainability in Predictive Modelling....6

6.4 Current Research Gap....6

7. Methodology....7

7.1 Research Design....7

7.2 Data Collection and Preprocessing....7

7.3 Model Development....7

7.4 Model Evaluation and Interpretability....8

8. Expected Outcomes....8

9. Project Plan and Timeline....8

10. Ethical Considerations....9

11. Risk Assessment and Mitigation....9

12. Resources Required....10

13. Contribution to Knowledge....10

14. Conclusion....10

Reference....12

1. Introduction

Customer behavior is based on understanding and predicting the behavior of different customer clusters, and this has become a major priority of business organizations working under a very competitive and dynamic market environment. Conventional customer analytics methods of assuming fixed segmentation, past sales analysis, or ad hoc predictions are progressively limited in describing the complexity and dynamism of the present consumer market. Adaptive machine learning is also a promising way to do so by making it possible to construct predictive models that utilize new data and learn in response to alterations in behavior patterns. With the combination of explainable AI methods, like SHAP and LIME, companies will have clear information on the trends worldwide and each specific individual prediction and be able to use this information to create personal approaches to market shifts.

The study also seeks to come up with adaptive and interpretable machine learning models capable of predicting customer behavior effectively with actionable explanations to the decision-makers. These models will enable individual business strategies, as they will identify the factors that influence them, long-term forecasts, and clear explanations of the forecasts. Predictions interpretation capability is essential to operationalizing machine learning knowledge in marketing, product development, and customer interaction to ensure that the strategy formulation process is a data-driven, dynamic formulation and geared towards organizational objectives.

2. Background and Rationale

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